Unit 19 The A/A Test 200

(1) Why A/A Tests? 200

X (1) What is an A/A Test? X

→ An A/A test is when you split traffic into two groups (A and A), but you give both groups the exact same experience - there is no treatment.

→ So, you're testing the system itself - not a feature.

X (2) Why Run an A/A Test? X

1. To validate the Experimentation System.

Imagine you're running a real A/B test, and your metrics show a statistically significant lift.
But what if the lift is due to a bug in your logging or randomization?
An A/A test should not show any significant - if it does, something wrong.

Analogy:
Think of A/A testing like a calibration test in a lab
Before weighing something, you put two weights that are supposed to be identical on the scale. If the scale shows different weights - the scale is broken.

2. To Estimate Type I Error (False Positives)

Let's say we do ~~A~~ 100 A/A tests.

Each test compares two identical groups, so under the null hypothesis (H_0), the true difference is 0

If your significance level is $\alpha = 0.05$, then you expect 5% of tests to show a false positive by random chance.

If you see 10 out of 100 A/A tests show significance, then your actual Type I error is 0.10, which is twice as high as expected — your system is too sensitive, which is dangerous

This connects to:

- [x] Multiple Testing (Unit 8)
- [x] Family-Wise Error Rate (FWER)
- [x] False Discovery Rate (FDR).

3. To understand Statistical Noise

Even with a well-calibrated system, real-world randomness can cause fluctuations.

A/A test help estimate variance and metric sensitivity so you can:

- Set appropriate thresholds for significance
- Understand how noisy your metric are.
- Diagnose issues with underpowered tests (low sample size, high variance)

X (3) Real-World Analogy: Doctor Testing a Thermometer X

Say a doctor uses a thermometer on the same person twice, 5 minutes apart:

- If the reading are drastically different, she knows the thermometer is broken,
- If they're slightly off, she can estimate the device's noise or tolerance.

An A/A test is exactly this: test the measurement tool (the experiment framework), not the medicine (treatment)

(2) How to Run A/A Tests: Core Principles:

1. Randomly Split Users into Two Groups.

- Use the same randomization infrastructure that you'd use in
- Assign users GroupA1 and GroupA2 - both groups see identical experiences.

The 3 validations

↳ Randomization logic
↳ Bucketing logic
↳ Data collection consistency.

> You can repeat A/A tests multiple times to ensure the system behaves stably over time.

2. Run at Production Scale

Dont just run a toy test on 1% of users

⇨ Run the A/A test at full production scale, just like a real experiment.

- Real traffic pattern
- Real sessions
- Real devices.

Why? This reveals real-world issues: time-based bias, caching bugs, out-of-memory errors, sampling bias etc.

3. Run Multiple A/A Tests to check Type I Error Rate.

Let's say $\alpha = 0.05$. If you run 100 A/A tests, you expect ~5 false positives just by chance. If you're getting 10-15 significant results, your system is too noisy or broken.

This helps empirically validate your Type I error control.

This directly connects to:

- [x] Multiple Testing
- [x] P-value Distribution (should be uniform)
- [x] FWER and FDR Calibration.

4. Test Across Key Metrics

Don't test just a single metric (e.g. click-through-rate).

Instead:

- Run A/A for all key metrics used in experiments
- Compare both means and variance estimates.

This ensures that:

- Metrics are stable across groups.
- You're not under- or overestimating variance, which affects confidence intervals and p-values.

5. Check Statistical Properties.

For each A/A test:

- look at the distribution of p-values — it should be close to Uniform(0, 1) under the null.

- Analyze the variance of metrics across groups — they should be similar.
- Use QQ plots to compare metric distributions.
- Use histograms of p-values and effect sizes.

If any of these show systematic bias (eg. shift in mean, skewed p-value distribution), your experiment infra may be broken.

(6) Real-World Analogy: Testing a Scale in a Lab.

Imagine a lab has a digital scale. Before weighing medicine, they:

1. Place identical weights on it (A vs A)
2. Repeat the process 100 times.
3. Make sure difference in readings is small and randomly varies around 0.
4. If the scale keeps showing "weight A is heavier", they know it's biased or miscalibrated.

Similarly, A/A tests test the testing system itself.

(7) Summary Table: How to Run A/A Tests

Step	What it Validates
Random split (A1 vs A2)	Randomization, hashing
Production-scale test	Real-world performance & edge cases
Run many A/A tests	Type I error calibration
Test multiple metrics	Metric stability, sensitivity
Check p-value distribution	Correct null hypothesis behavior
Analyze variance	CI correctness
Consistency	

(3) What Does it Mean when an A/A Test fails?

An A/A test fails when it detects a statistically significant difference between two identical ~~expression~~ experiences.

Since both groups receive exactly the same treatment, any statistically significant difference suggests a problem with the experiment system - not the treatment (because there is none).

Key Failure Scenarios in A/A Tests

1. Too Many False Positives (inflated Type I Error Rate)

→ Expected: if $\alpha = 0.05$, ~5% of tests should be falsely significant.

→ Failure: You see 10%, 15%, or more significant tests ⇒ The system is too sensitive.

Causes:

- Incorrect p-value calculations
- Underestimated variance
- Biased randomization
- Multiple testing without Correction (e.g. Bonferroni)

This directly overlaps with Unit 18

- Multiple Testing
- FWER (Family-Wise Error Rate)
- False Discovery Rate (FDR)

2. Inconsistent or Biased Randomization

Random assignment must be:

- Independant
- Uniformely distributed
- Sticky (same user always get same buc

Failure: Certain groups (e.g. mobile vs desktop) are unevenly distributed.

Effect: Differances in behavior arise due to group composition, not the experiment.

Example:

- A1 = 60% mobile users
- A2 = 40% mobile users

→ Differance in click rates isn't due to treatment but group skew.

3. Instrumentation or Logging Bugs

If metric are:

- Logged differently across buckets
- Missing events
- Incorrectly aggregated.

Then even identical groups can show different results

| Failure shows up as:

- Effect size ≠ 0 (non-zero mean diff)
- Influenced by subtleties
- p-values too small.

4. Time-Based Bias

You ran A1 in the morning and A2 in the evening
→ Different user behavior based on time, not treatment.

| This is often called temporal bias.

5. Cross-Bucket Contamination.

A user sees experience from both A1 and A2 (e.g. due to not using a sticky ID)
Result: Mixed experiences, affecting measurement accuracy.

6 Real-World Analogy: Faulty Thermometer Test:

Let's say you're checking if a thermometer is working by measuring the same room twice.
Expected: Both readings ≈ same

Failure: One shows 22.1°C, another shows 24.5°C
→ Thermometer is faulty or biased same logic applies in A/A testing.

7 Summary Table: Common A/A Failures

Failure Type	Symptom	Root Cause Example
Too many false positives	75% tests significant at $\alpha = 0.05$	Wrong p-value logic/ poor variance
Group imbalance	Unequal user types across buckets	Broken or biased randomization

Metric inconsistency	Different means in identical flows	Logging bugs, missed events.
Time-based bias	Diurnal behavior differences	Not controlling test time.
Contamination	Shared experience across buckets	User ID not sticky.

How to Debug A/A Failures

→ Check bucket distributions by user / device / location.

→ Plot p-value histogram. Should be Uniform (0, 1)

→ Compare variance estimates between buckets.

→ Run A/A test sequentially and compute actual false positive rate

→ Visualize metric over time to detect time-based bias.

dhawalchoriya